

Large Language Models and grammatical Overcorrection

Qiancao Jiang

LLMs as models of human sentence processing · SoSe 2026
Prof. Vera Demberg · Universität des Saarlandes

Motivation

LLMs tend to completely rewrite sentences written by non-native speakers to conform to the expression style of native speakers (Kopylova et al., 2026; Munda et al., 2024). In contrast, BabyLM (Warstadt et al., 2023) has lower cognitive complexity, and I hypothesize that it may align more naturally with the “minimal correction” teaching principle, thereby avoiding “overcorrection.”

Research question

However, the experimental results involving BabyLMs indicate that the non-instruction-tuned models are unable to perform this grammatical error correction task. Consequently, my research question shifted to: **Does the number of parameters in locally deployable, open-source, and free large language models affect the phenomenon of overcorrection when the models correct grammatical errors?**

Hypothesis

I treat overcorrection as the normalized edit distance plus the number of FP. My hypothesis is that large language models with fewer parameters are less prone to overcorrection.

Method

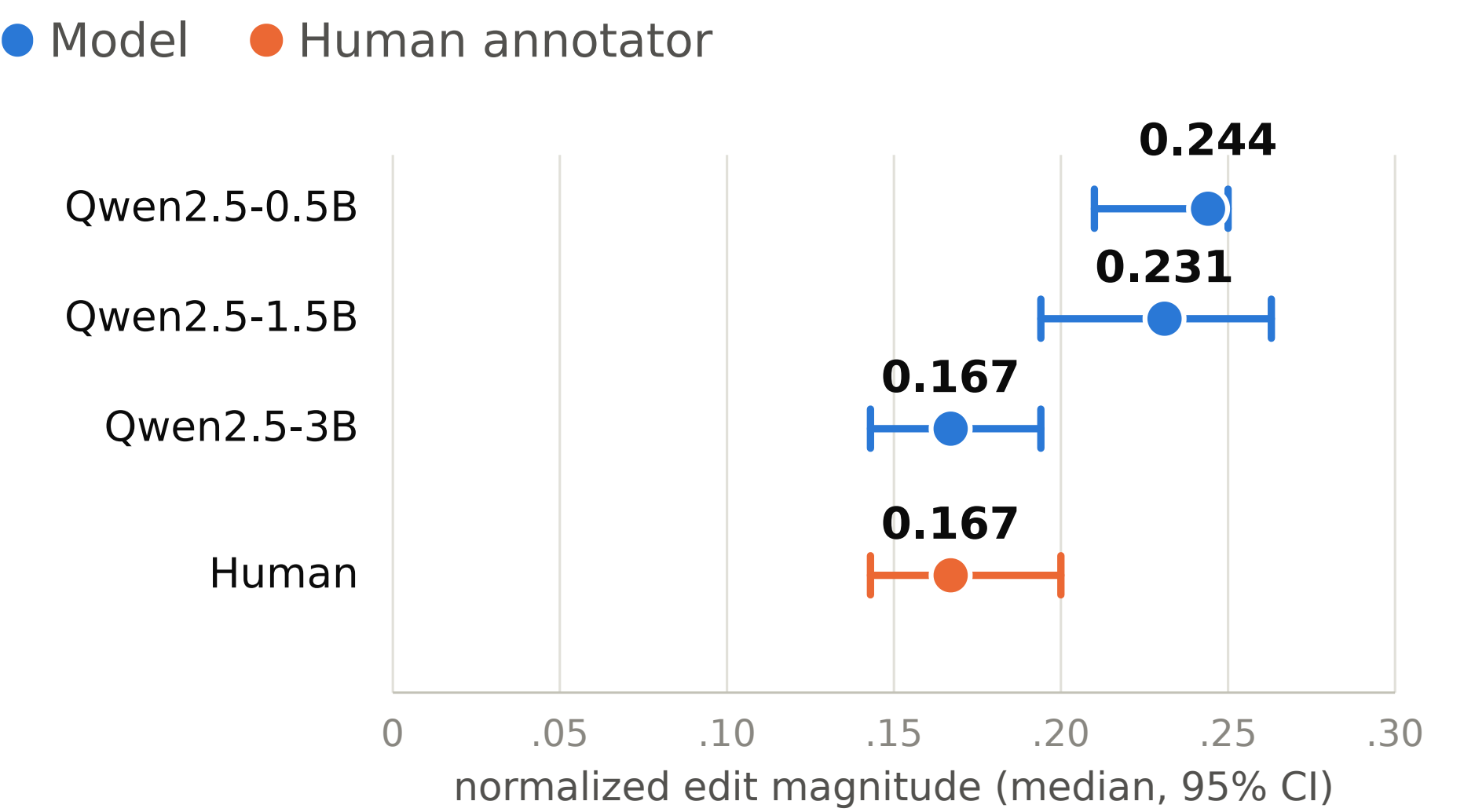
Corpus	W&I+LOCNESS v2.1 (BEA-2019), CEFR-A dev split
Sampling	1,037 → 705 after filtering → 200 items, seed 42
Filters	5–40 tokens; ≥1 non-noop edit; no punctuation- or orthography-only items; 14 degenerate items dropped
Conditions	Qwen2.5-0.5B / 1.5B / 3B-Instruct — same family, same tokenizer, only parameter count varies ; human annotator as a fourth reference point
Prompt	zero-shot: “Correct only grammatical errors in this sentence and then only output the corrected sentence.”
Decoding	greedy; 200 × 3 = 600 generations, 0 failures
Post-processing	model commentary removed by manual annotation (prefaces 7 / 15 / 0; trailing comments 0 / 2 / 5), archived in the repository
Measures	normalized word-level edit magnitude vs the source; ERRANT span-based P / R / F _{0.5} vs the gold correction
Statistics	Friedman (tie-corrected); post-hoc Wilcoxon signed-rank, Holm-corrected; rank-biserial <i>r</i> ; 4,000-resample percentile bootstrap CIs

Why two measures

Edit magnitude asks **how much** a condition changed the sentence; ERRANT asks **whether the changes were the right ones**. Magnitude alone cannot separate a model that edits little because it is restrained from one that edits little because it cannot correct — a model that simply copies the input scores a perfect 0.

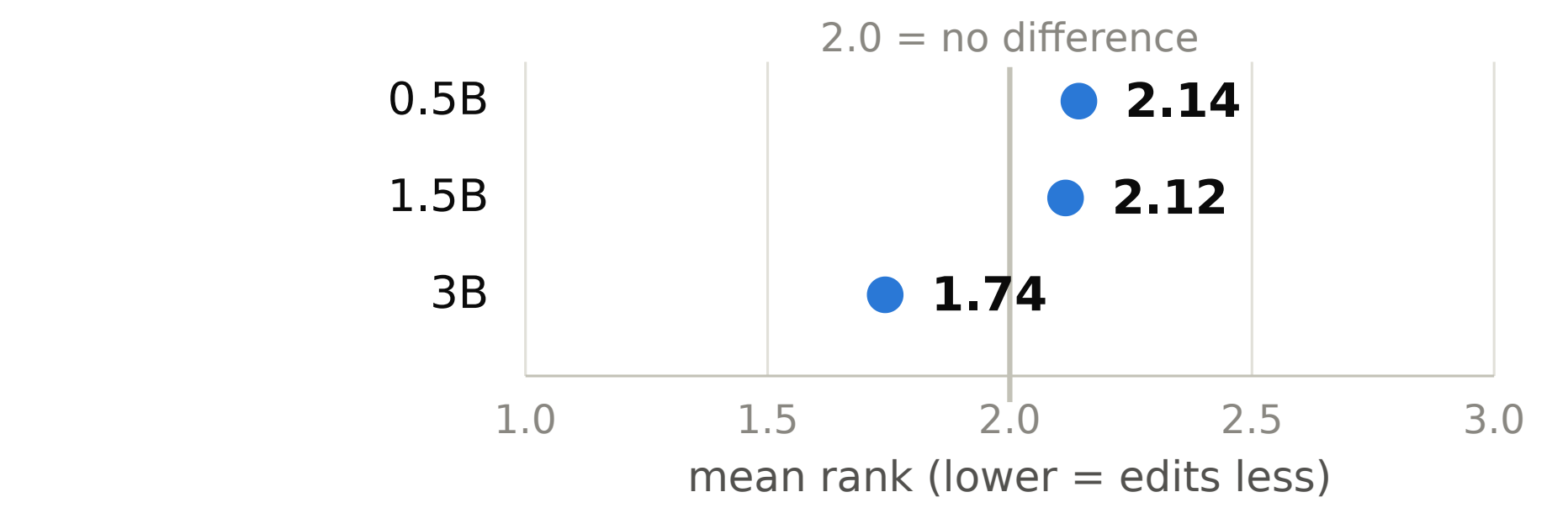
Finding 1 · a step, not a gradient

As the number of parameters increases, the model becomes less prone to overcorrection; however, this decrease follows a “stepwise” rather than a “gradual” pattern. There is a threshold between 1.5B and 3B, rather than a continuous pattern where “the more parameters, the fewer corrections.”



3B and the human annotator have the same median and are statistically indistinguishable ($p_{\text{holm}} = .57$, $r = .09$), while 0.5B and 1.5B do not differ from each other ($p_{\text{holm}} = .57$, $r = .10$) and both edit far more than the annotator ($r = .50$ / $.46$).

Mean rank within each sentence



Friedman $\chi^2(2) = 25.64$, $p < .001$. **0.5B and 1.5B are nearly tied; 3B sits clearly below both** ($p < .001$, $r = .53$ / $.56$) — the shape of a step rather than a gradient.

One sentence, four conditions

source	It 's difficult answer at the question " what are you going to do in the future ? " if the only one who has to know it is in two minds .	edits
human	It 's difficult to answer the question " what are you going to do in the future ? " if the only one who has to know it is in two minds .	2 · .061
3B	It's difficult to answer the question "What are you going to do in the future?" if the only one who has to know it is in two minds.	2 · .061
1.5B	It's difficult to answer the question "What are you going to do in the future?" if the person who knows must be in two minds.	4 · .273
0.5B	It's challenging to answer the question "What are you going to do in the future? " when the only one who knows it is unsure .	5 · .303

Human makes exactly two minimal edits (insert to, delete at). **3B makes the same two** — its output is token-for-token the annotator's. 1.5B rewrites the subordinate clause. 0.5B also replaces difficult, if and in two minds — none of which contained a grammatical error.

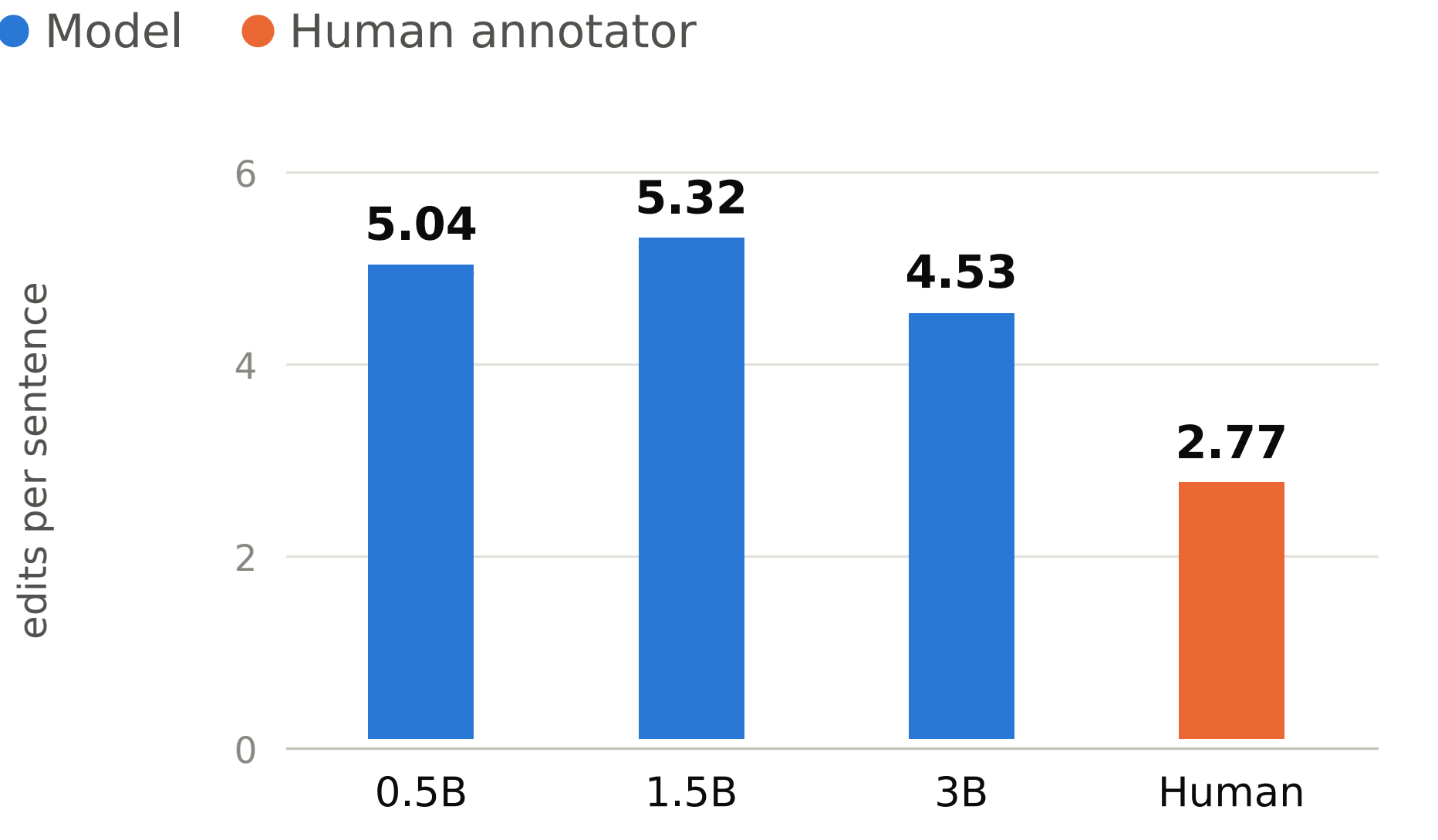
Finding 2 · more fixed, less broken

ERRANT span-based scores against the gold correction (200 items, 553 gold edits)

Model	TP	FP	FN	P	R	F _{0.5}	F _{0.5} no ORTH
0.5B	121	887	432	.120	.219	.132	.161
1.5B	163	900	390	.153	.295	.170	.208
3B	181	724	372	.200	.327	.217	.289

While fixing more actual errors, 3B made fewer unnecessary changes. Compared to 0.5B, 3B shows an increase in TP and a decrease in FP. This indicates that 60 more correct changes were made and 163 fewer unnecessary changes were made. Most of its changes are still unnecessary (precision 0.20).

How much gets changed per sentence



Every model rewrites far more of each sentence than the annotator does. **Even 3B changes 1.6× as many spans as the human**, and its precision stays at .20 — most of what it changes, a human would have left alone.

Finding 3 · ORTH is an artefact

The ORTH category — casing and whitespace only

Model	TP	FP	FN
0.5B	7	251	12
1.5B	7	251	12
3B	7	278	12

TP and FN are **identical** across all three conditions and FP varies by 11%, while total FP varies by 24%. This category tracks output formatting, not correction behaviour. Excluding it widens 3B's false-positive advantage over 0.5B from **18% to 30%**.

Conclusion

Based on the experimental results, I can reject my hypothesis; in fact, large language models with more parameters are less prone to overcorrection. When selecting large language models to assist with English instruction, educational institutions and individual teachers should, whenever possible, choose free large language models with a high number of parameters that can be deployed on local computers.

3B ≈ human edit magnitude: no detectable difference ($r = .09$)	0.5B ≈ 1.5B no detectable difference either ($r = .10$)
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References

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Code & materials

<https://github.com/whks-wu/llm-seminar-project.git>

Corpus text is not redistributed (W&I+LOCNESS licence §6). The repository ships the sampling manifest (ids + seed + checksum), the manual annotation of model commentary, all analysis code, and the per-item measurement table.